

Assignment 1: Summary Report

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Executive Summary

I performed a series of experiments on 14 versions of a neural network that could classify IMDB movie reviews as positive or negative. The model learned from the class was already decent, and it was correctly labeling approximately 88% of the reviews it has not seen. It was hardly affected when the size or shape of the network was changed. All versions achieved between 88.8% and 89.3% on the validation reviews, and the difference is probably due to chance. That one thing that mattered was overfitting. When given no assistance, after just 2 to 4 epochs, all of the models began to memorize their training reviews and thereafter their accuracy on unpresented reviews was found to be in a downward spiral. This was hindered most by drop-out. My recommendation is the original network I created in class (16x2 network) with 50% dropout. It was the top test accuracy among all 14 models at 88.59% and it performed decently even with the longer training periods compared to its actual requirement.

Key Terms

Term	What it means here
Training, validation and test sets	There are 50,000 reviews, which are evenly divided between the 3. There are 50,000 reviews evenly divided between the 3. It's trained from 15,000 reviews. Additional 10,000 validation reviews for comparison and selection of a winner. The final 25,000 test reviews are not modified until the end, and are meant as a fair test on reviews the model has never encountered.
Accuracy	The percentage of review items correctly labeled. The data is split roughly half and half between positive and negative, so that a random guesser can make about 50% of his guesses.
Epoch	A full lap of the training reviews is conducted. All of the following models were trained for 20 epochs.
Hidden layers and units	The in-between layers between the input and the answer and how many neurons are there in each of these layers. The more that are added, the better the model will be able to learn more complex patterns, but also more rote-learned.
Overfitting	A model that is just getting worse on new data as it is getting better on training data due to overfitting. I use the "gap" between validation and training accuracy after 20 epochs as a measurement.
Loss function	The difference between the model's prediction and the actual data that the model is learning from. I did compare BCE which is the typical choice for problems that are binary in nature, with mean squared error (MSE).
Activation function	The simple rule that each neuron adopts to its input. The old default was tanh, but today it is ReLU.
Dropout and L2	For some, there are two ways to discourage memorizing. Dropout randomly turns off half of the neurons each training step. L2 will punish the model for using very large weights.

1. How I Ran the Tests

Baseline: Two hidden layers of 16 ReLU (trained with binary crossentropy) units from class. All reviews are turned into a list of 10,000 most frequent words present in those reviews. I then constructed 13 variations off of that. One will affect one setting and the other will affect a different setting, thus any differences in results can be attributed to the setting that was different. The same data set, the same random initialisation and the same 20 epochs of training were used for all 14 models.

For each model I recorded its best validation accuracy and the epoch where it happened. I then took the model as it was at that best epoch and scored it on the test set. I picked winners using validation results only. The test set never influenced a decision.

2. What I Found

Number of layers. The one hidden layer obtained 89.04% on validation, the two hidden layers 88.95% and the three hidden layers 88.78%. Going deeper didn't help. The three layer network reached its peak first and then the line dropped off the fastest, so it was primarily more space for memorization. A shallow network can suffice for this type of input.

Number of units. I tried 8, 16, 32 and 64 units per layer. Validation accuracy was between 88.81% and 89.26% (64 units at the top). I don't take this lead very seriously, however, because that model started to fade after just 2 epochs, and also had a lot of fluctuation in the validation scores between epochs. The overfitting was barely affected by the width. The difference for all four sizes was between 12.5 and 12.8 points.

Loss function. The mean squared error achieved a score of 89.01% while the binary crossentropy achieved 88.95% not much away from normal run-to-run variation. It performed a bit better due to its slight under-fitting (by 11.7 points, not 12.8), likely because it's being less aggressive on predictions it already has a high degree of confidence in. If I were to still have binary crossentropy, I would still hold it. It is meant for binary probabilities and punishes incorrect answers with a higher severity for surety. At 16 and 32 units, activation function tanh outperformed ReLU in terms of peak accuracy. It lagged behind in the stability category. Both models with tanh had the two highest overfitting ranges during all epochs of the study, and the lowest test accuracy after all 20 epochs (about 85.2%). The default is ReLU.

Dropout and L2. The only modification that was clearly evident in changing the behavior of the model is the change of this variable. The baseline network had a 50% drop out rate, and the validation result was 89.25% at the best epoch, which was 8 when it improved for two times longer than 4. The results showed that its overfitting gap reduced from 12.8 to 8.1 points while its accuracy on test set was highest at 88.59% for all models. It also was very resilient to over training, as after 20 epochs, it retained a test score of 87.64% whilst the baseline had fallen to 85.84%. The combined L2 did not improve accuracy, but narrowed the lead to 10.8 points. The second smallest gap was 9.6 points with drop-out and L2 sharing the 64-unit network

3. Results at a Glance

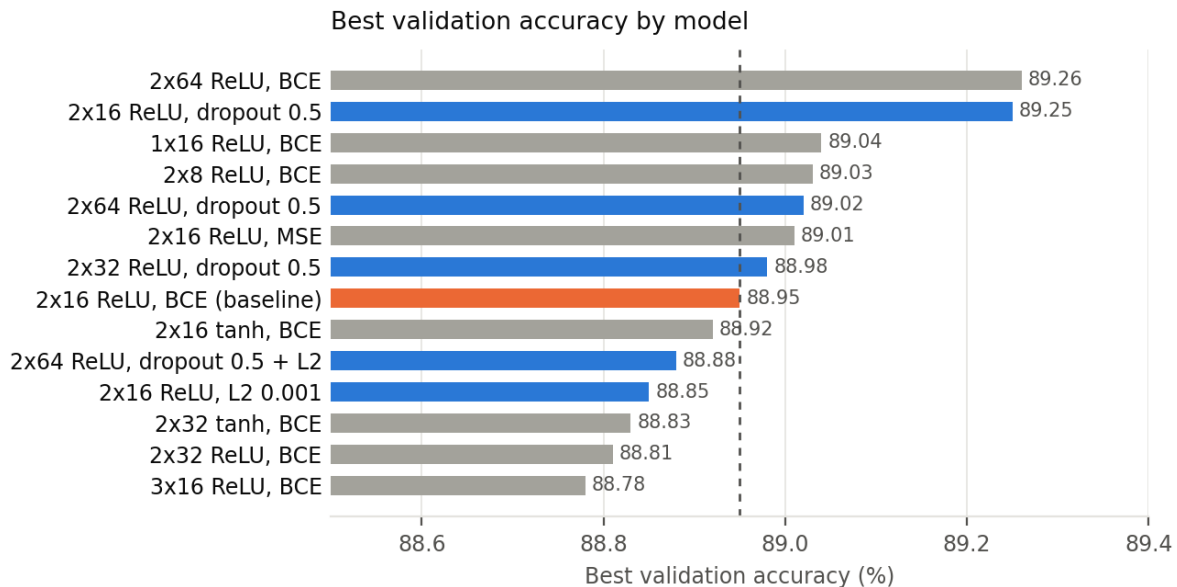
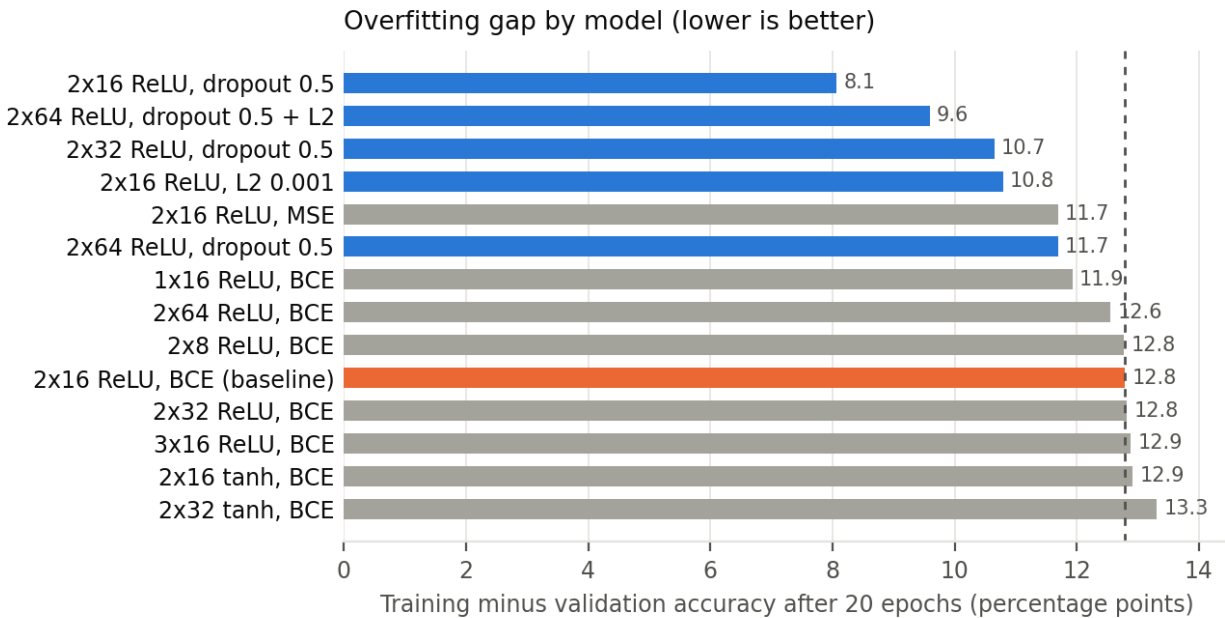


Figure 1. Best validation accuracy for each model. The whole range is under half a percentage point. Orange marks the baseline (also the dashed line) and blue marks models with dropout or L2.

Table 1. All 14 models, ranked by best validation accuracy. “2x16” means two hidden layers with 16 units each. Test accuracy uses each model at its best epoch. Overfitting gap is in percentage points after 20 epochs (lower is better). Blue row: recommended model. Grey row: baseline.

Rank	Model	Change tested	Validation acc.	Test acc.	Overfitting gap
1	2x64 ReLU, BCE	Width	89.26%	88.24%	12.6
2	2x16 ReLU, dropout 0.5	Regularization	89.25%	88.59%	8.1
3	1x16 ReLU, BCE	Depth	89.04%	88.32%	11.9
4	2x8 ReLU, BCE	Width	89.03%	88.11%	12.8
5	2x64 ReLU, dropout 0.5	Regularization	89.02%	88.16%	11.7
6	2x16 ReLU, MSE	Loss function	89.01%	88.34%	11.7
7	2x32 ReLU, dropout 0.5	Regularization	88.98%	88.40%	10.7
8	2x16 ReLU, BCE (baseline)	Baseline	88.95%	88.21%	12.8
9	2x16 tanh, BCE	Activation	88.92%	88.18%	12.9
10	2x64 ReLU, dropout 0.5 + L2	Regularization	88.88%	88.14%	9.6
11	2x16 ReLU, L2 0.001	Regularization	88.85%	88.27%	10.8
12	2x32 tanh, BCE	Activation	88.83%	88.21%	13.3
13	2x32 ReLU, BCE	Width	88.81%	88.25%	12.8
14	3x16 ReLU, BCE	Depth	88.78%	88.02%	12.9



4. Recommendation

Had I just looked at the highest validation score, I'd have chosen the 64 unit network (just 0.01 points ahead of the dropout model). I marked that off in my notebook by retraining the 64 unit model with all 25,000 training reviews (for 2 epochs) and it had reached its best performance. It achieved a test set score of 86.62% compared to the baseline score of 88.21%. If you have a 0.01 point lead on one sample set, it's noise. And if the model peaks after two epochs, there isn't a lot of margin when anything changes in training. I suggest to use 2x16 ReLU network with dropout rate of 50% instead. It has the best validation score tied with the smallest overfitting gap, and it doesn't require the exact right moment to stop training. In practice I would combine it with early stopping, which stops training automatically when the accuracy on the validation set doesn't improve.

5. Limitations and Next Steps

One training was conducted for each model using one random seed. There are only a few tenths of a percent difference between most of the models, and the difference from a different seed alone could be that much so the middle of the ranking should not be interpreted as being made up of the same model. It would give more confidence if each model was run with multiple seeds and the results averaged. My next steps would be to retrain the recommended dropout model on all 25,000 training reviews to check its test score and to test dropout rates other than 50%.

6. Conclusion

The network from class was almost at its optimum performance with this setup. None of the changes to the accuracy were more than half a point when the depth or width of the network were increased or the loss function or activation function changed. The true challenge was over fitting. The tools that worked were those that were directed directly at it: stopping at the best epoch and dropout. In this problem, a small network with dropout is best suited for the problem to be both accurate and reliable.